Information Retrieval and Extraction : Lecture V (IRE Evaluation Systems)

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Outline I

1. Information Retrieval Evaluation Systems
   - Creating Ground-truth Dataset
   - Precision-Recall Trade-off
   - Measuring Relevance
   - User based Metrics

Information Retrieval Evaluation Systems

Test Reference Collection

- Test Reference Collection
  - A collection of documents
  - A set of example information requests (queries)
  - A set of relevant documents for each information request (or a relevance judgment for each document-query pair)
- Evaluation Measure
  - Qualify the similarity between the set of documents retrieved
    - set of relevant documents provided by the specialists (assessors)
  - Provide an estimation of the goodness of the retrieval strategy

Information Retrieval Evaluation Systems

Understanding

- User submits a query and receives an answer back
  - quality of the generated answer set
- User specifies his information need *iteratively*
  - user effort, interface design, system's guidance, session duration, or the context in which the query is posed
- Recall : The fraction of the relevant documents which has been retrieved
  - assume that all the documents in the answer set have been examined
- Precision : The fraction of the retrieved documents which is relevant

Information Retrieval Evaluation Systems

Creating Ground-truth Dataset

- The test document collection and suite of information needs have to be of a reasonable size:
  - 50 information needs has usually been found to be a sufficient minimum
  - Ex: 200 news articles was accepted as a good number
- Relevance is assessed relative to an information need
  - not a query

Summary of Insights

- *Using ChatGPT at an earlier age seems to have an adverse impact on development of cognitive abilities of a kid?*
  - A document is relevant if it addresses the stated information need
  - Not because it just happens to contain all the words in the query

Information Retrieval Evaluation Systems

Creating Ground-truth Dataset

- Test collections are available to gauge the performance of your IRE algorithm/model!
  - How to create them?
- Given information needs and documents, you need to collect relevance assessments
  - Almost impossible to do for a large dataset
  - Solution : Pooling
    - where relevance is assessed over a subset of the collection
    - by combining different annotators
    - based on majority
    - mutual agreement
    - or, *Kohen Kappa* co-efficient

Information Retrieval Evaluation Systems

Creating Ground-truth Dataset

A1/A2	Yes	No	Total
Yes	300	20	320
No	10	70	80
Total	310	90	400

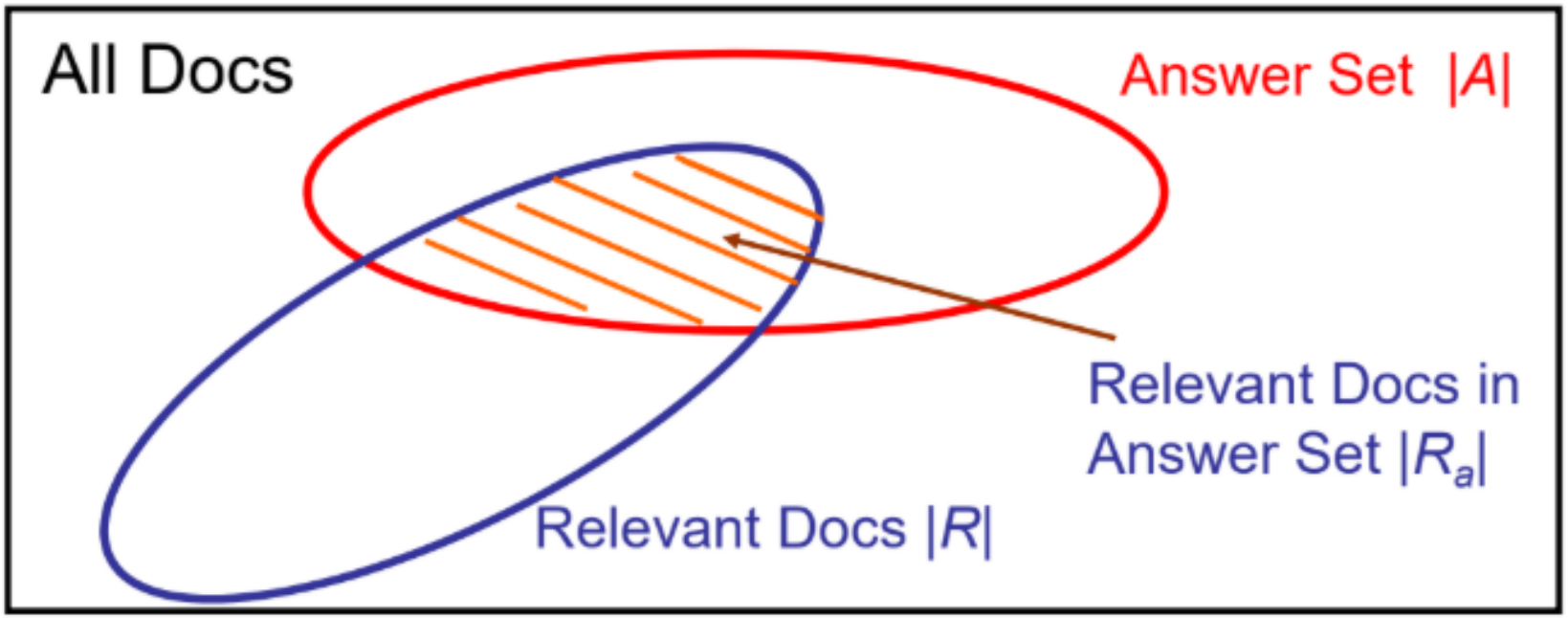
Information Retrieval Evaluation Systems

Kohen Kappa co-efficient

- Observed proportion of the times the annotators agreed
  - P(A)= $\frac{(300+70)}{400} = 0.925$
- Observed Classwise
  - P(nonrelevant) $= \frac{80}{400} * \frac{90}{400}$
  - P(relevant) $= \frac{320}{400} * \frac{310}{400}$
- Probability that the two judges agreed by chance, P(E)= $P(nonrelevant) + P(relevant)$
- Kohen Kappa, $\frac{(P(A)-P(E))}{(1-P(E))}$
  - If greater than 0.8, it is very good
  - If within $0.67 - 0.8$, it is still fair

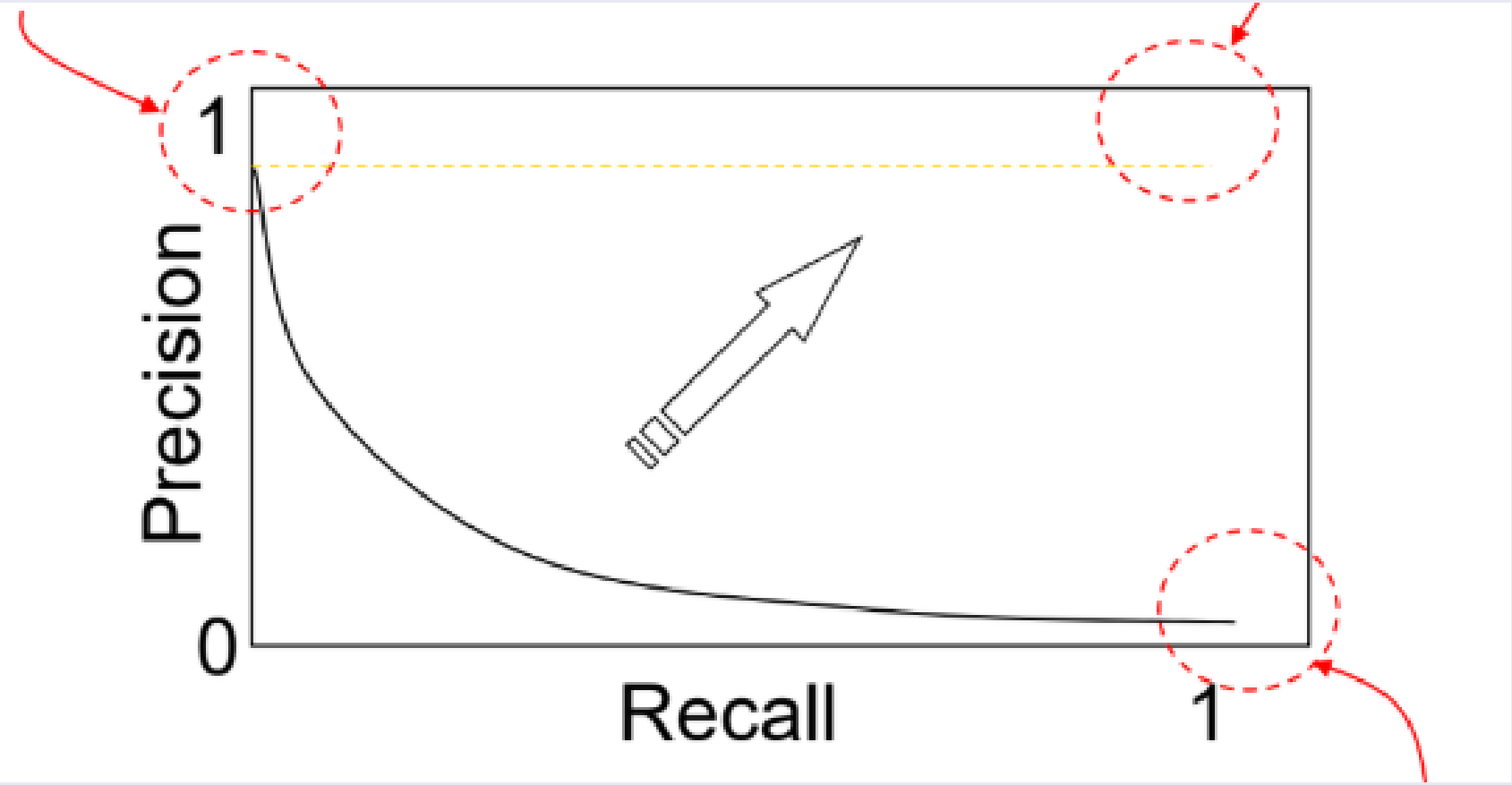
Information Retrieval Evaluation Systems

Representation



Information Retrieval Evaluation Systems

Precision-Recall Trade-off



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Precision-Recall Trade-off

Cm	Rel	NRel
Ret	tp	fp
NRet	fn	tn

Recap

- Precision (P) is the fraction of retrieved documents that are relevant
- Recall (R) is the fraction of relevant documents that are retrieved
- Accuracy (A) is the fraction of its classifications that are correct
  - data is extremely skewed, increasing accuracy does not ensure

Information Retrieval Evaluation Systems

Example

- An IR system returns 2230 relevant tweets, and 1110 non-relevant tweets related to *Epstein Files* when used this specific hashtag over last 3 days. There are a total of 2200000 tweets that were published in last 3 days.
  - What is the *precision*?
  - How is the *recall*?
  - Any suggestions?

Summary of Insights

- In a ranked retrieval context,
  - Higher the rank, more the relevance
- if the $(k+1)^{th}$ document retrieved is non-relevant,
  - then, has my recall dropped since k^{th} retrieval?
  - what about precision?

Information Retrieval Evaluation Systems

Average Precision

- Average Precision at a given document cutoff values
  - Compute the average precision
    - when top 5, 10, 15, 20,30, 50 or 100 relevant documents have been seen
  - Focus on how well the system ranks the top-k documents
    - Provide additional information on the retrieval performance
    - Take (weighted) average over results

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Average Precision

1. d_{123} • (P=1.0)	6. d_9 • (P=0.5)	11. d_{38}
2. d_{84}	7. d_{511}	12. d_{48}
3. d_{56} • (P=0.66)	8. d_{129}	13. d_{250}
4. d_6	9. d_{187}	14. d_{113}
5. d_8	10. d_{25} • (P=0.4)	15. d_3 • (P=0.3)

(1.0+0.66+0.5+0.4+0.3)/5=0.57

Average Precision

- It favors systems which retrieve relevant docs quickly (early in the ranking)
- this measure depends heavily on highly ranked relevant documents
  - Also, referred as non-interpolated average precision
- MAP is the AP over all possible queries

Information Retrieval Evaluation Systems

R-Precision

- generate a single value summary of ranking
- by computing the precision at the R-th position in the ranking
  - R is the total number of relevant docs for the current query

R-Precision

1. d_{123} ●
2. d_{84}
3. d_{56} ● ■
4. d_{6}
5. d_{8}
6. d_{9} ●
7. d_{511}
8. d_{129} ■
9. d_{187}
10. d_{25} ●
11. d_{38}
12. d_{48}
13. d_{250}
14. d_{113}
15. d_{3} ● ■

$R_q=\{d_3, d_5, d_9, d_{25}, d_{39}, d_{44}, d_{56}, d_{71}, d_{89}, d_{123}\}$

•10 relevant documents (●)

=> *R*-precision = 4/10=0.4

$R_q=\{d_3, d_{56}, d_{129}\}$

•3 relevant document (■)

=>*R*-precision=1/3=0.33

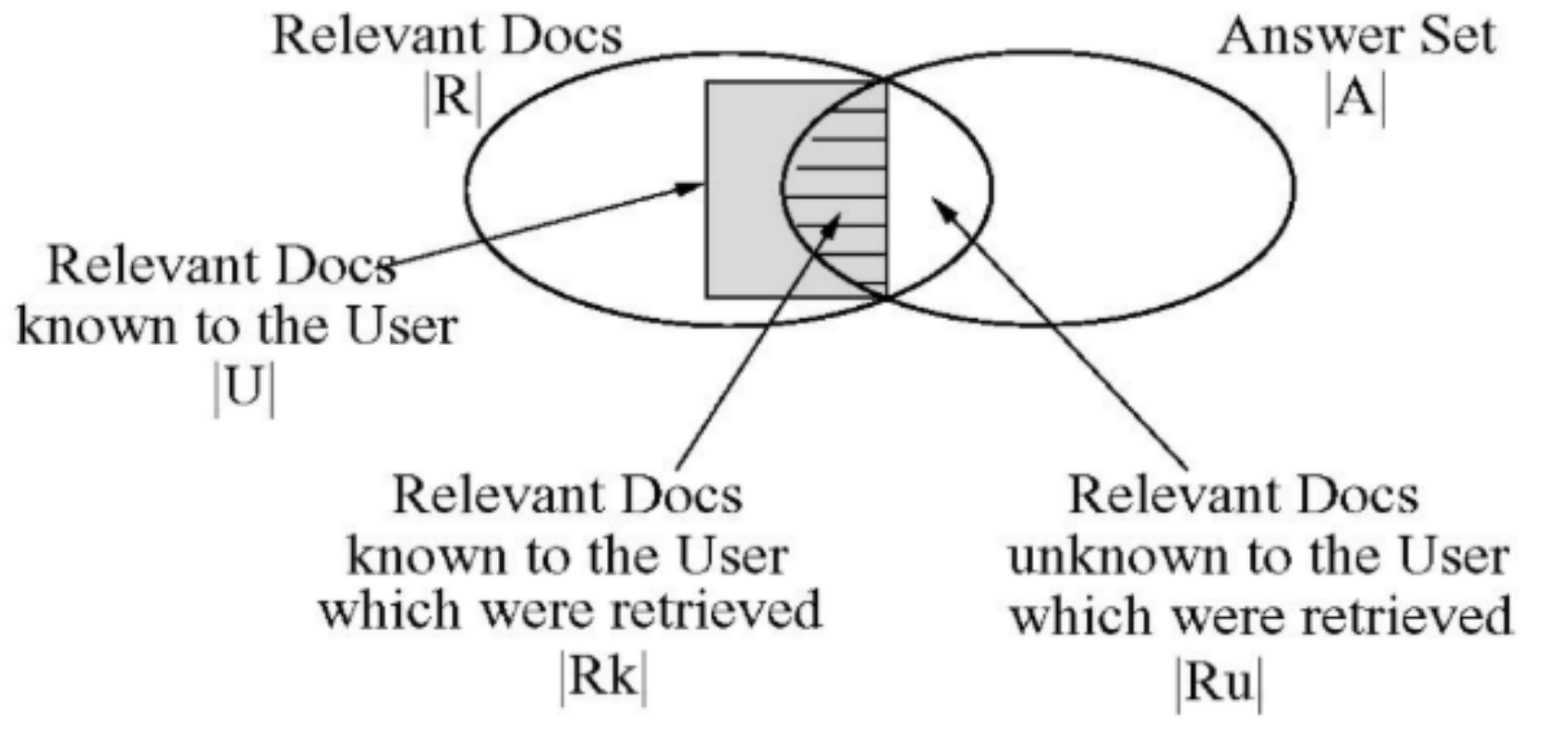
Information Retrieval Evaluation Systems

R-Precision and MAP : Insights

- Assumption
  - Set of relevant documents for a query is the same, independent of the user
  - Users might have a different interpretation of document relevance
- Other Measures
  - Coverage ratio
  - Novelty ratio
  - Relative recall
  - Recall effect

Measuring Relevance

User based Metrics



Measuring Relevance

User based Metrics

- Coverage ratio, $\frac{|Rk|}{|U|}$
  - The fraction of relevant docs known to the user which has been retrieved
  - High $\rightarrow$find most of the relevant docs user expected to see
- Novelty ratio, $\frac{|Ru|}{|Ru|+|Rk|}$
  - The fraction of relevant docs retrieved which is unknown to the user
  - High $\rightarrow$find (reveal) many new relevant docs (information) the user previously unknown

References

Introduction to Information Retrieval by Christopher Manning (ch-8)

- Only the sub topics covered in slides/class

Thank you!